Christof Teuscher

ECE 410/510: Hardware for AI and ML, Spring 2026

Codefest #4: Quantization + INT8 unit in HDL

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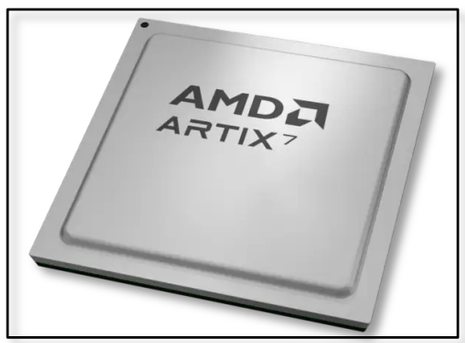

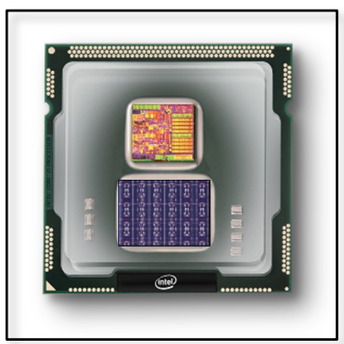

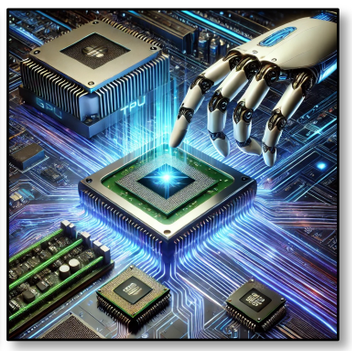

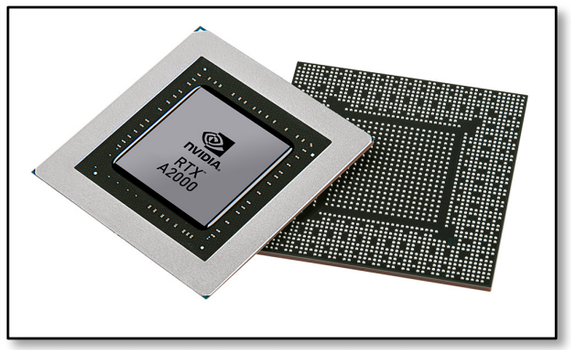

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Today

CMAN

Manual INT8 symmetric quantization.

Using a spreadsheet or calculator is fine!

CLLM

LLM-assisted HDL: INT8 MAC unit generation and review.

See VLSI lectures slides for tools. Make your workflows friction-free so you can iterate and gradually build something more complex.

COPT

cocotb testbench and your project compute core.

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Claudian Your friendly ECE 410/510 Hardware for AI and ML TA!

teuscher@pdx.edu Sign out

Codefest 3 Time remaining: **9:34**

Where do I start with CMAN?

Welcome! Great question to start with. CMAN is all about analyzing DRAM traffic for matrix multiplication, so let's make sure you have a solid foundation before diving into the calculations.

Here are a few warm-up questions to get you oriented:

- **What is the size of the matrices?** The problem says N×N where N=32. Each element is FP32 — how many bytes is that per element?
- **Think about the naive triple loop.** For a single output element C[i][j], you compute a dot product. How many multiplications and additions does that require? Which elements of A and B do you need to read?
- **"No data reuse"** is a key assumption in Task 1. What does that mean in terms of where every single memory access goes?

Start by thinking through Task 1 — just one output element C[i][j] first — before worrying about the full N×N output.

What's your initial thinking on how many times a single element of B gets accessed across the whole computation? 😊

All conversations are logged and may be reviewed for course improvement and analytical purposes.

Type your question... (Enter to send, Shift+Enter for new line) End Send

https://teuscher-lab.ece.pdx.edu/claudian

Presentations (1)

- **Goals**
  - Share your idea
  - Share your progress
  - Hone your presentation skills
- **Schedule**
  - Presentation duration: 1 minute (strict)
  - Schedule linked on Canvas
- **Instructions**
  - Create a single (= 1 = ONE) slide on which you answer the following questions:
    - What are you trying to do? I.e., what AI/ML algorithm/workload did you pick?
    - How have others implemented and/or accelerated this algorithm?
    - What are you doing differently/better/etc.?
    - What have you accomplished so far?
    - What will you do next and what remains to be done until you can declare success?
- Save the slide in the shared Google folder. URL on Canvas
- A strong presentation earns 1 token.

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Presentations (2)

Date	Week	Time	Name
Wed, April 22	4	4:55: PM	Eraganeni, Vamsidhar Reddy (Pronoun: He/Him)
Wed, April 22	4	4:57: PM	Kamarajugadda, Venkata Sriram (Pref: Venkata Sriram)
Wed, April 22	4	4:59: PM	Boss, David (Pronoun: He/Him)
Wed, April 22	4	5:01: PM	Hynes, Kay (Pref: Kay)
Wed, April 22	4	5:03: PM	Gilbert-Croysdale, Rebecca R. (Pronoun: She/Her)
Wed, April 22	4	5:05: PM	Charian, Sai Ganesh Reddy
Wed, April 29	5	4:55: PM	Vemula, Rajeshwar Reddy (Pronoun: he/him)
Wed, April 29	5	4:57: PM	Henrichs-Tarasenkov, Fyodor P. (Pref: Fedya, Pronoun: He/him)
Wed, April 29	5	4:59: PM	Herrera Moreno, Ivan (Pref: Ivan)
Wed, April 29	5	5:01: PM	Nguyen, Andrew (Pref: Bao)
Wed, April 29	5	5:03: PM	Chikani, Fatema Z. (Pronoun: She / Her)
Wed, April 29	5	5:05: PM	Munnangi, Pratibha (Pref: Pratibha, Pronoun: she/her)
Wed, May 6	6	4:55: PM	Ngo, Huy (Pref: Michael)
Wed, May 6	6	4:57: PM	Nguyen, Nhat M. (Pronoun: he/him)
Wed, May 6	6	4:59: PM	Naidu, Mahesh (Pref: Mahesh, Pronoun: He/Him)
Wed, May 6	6	5:01: PM	Rabiah, Yaqoub Y. (Pref: Yaqoub)
Wed, May 6	6	5:03: PM	Sabti, Saleh M. (Pref: Saleh, Pronoun: He/Him)
Wed, May 6	6	5:05: PM	Alali, Manaf A.
Wed, May 13	7	4:55: PM	Stanton, Andrew J. (Pref: Andrew, Pronoun: he/him)
Wed, May 13	7	4:57: PM	Bandla, Karthik
Wed, May 13	7	4:59: PM	Alrandi, Dawood A.
Wed, May 13	7	5:01: PM	Premarajan, Roshan B. (Pref: Roshan, Pronoun: hr/him)
Wed, May 13	7	5:03: PM	Vedantam, Venkata Krishna Kumar (Pronoun: His/Him)
Wed, May 13	7	5:05: PM	Esmaeil, Saleh A. (Pref: Sal, Pronoun: He/Him)
Wed, May 20	8	4:55: PM	Mills, Camden J. (Pref: Camden, Pronoun: He/Him)
Wed, May 20	8	4:57: PM	Day, Elliott F. (Pronoun: he/him)
Wed, May 20	8	4:59: PM	Mahajan, Ria (Pref: Ria)
Wed, May 20	8	5:01: PM	Khaddoori, Ibrahim A. (Pref: Ibrahim)
Wed, May 20	8	5:03: PM	AlQuzweni, Hussain A. (Pref: Hussain, Pronoun: he/him)
Wed, May 20	8	5:05: PM	Alotaibi, Mohammed M. (Pref: Mohammed)
Wed, May 27	9	4:55: PM	Wyse, Ian A. (Pronoun: he/him/his)
Wed, May 27	9	4:57: PM	Vanukuri, Sri Chandana Reddy
Wed, May 27	9	4:59: PM	Thakur, Amogh Vinod
Wed, May 27	9	5:01: PM	Kyatham, Vanisri (Pronoun: she/her)
Wed, May 27	9	5:03: PM	Vanamala, Naveen Babu (Pronoun: HE/HIM)
Wed, May 27	9	5:05: PM	Mallick, Devdutt S. (Pref: Devdutt, Pronoun: He/Him)
Wed, May 27	9	5:07: PM	Alshaiji, Mohammad A. (Pronoun: He/Him)
Wed, June 3	10	4:55: PM	Ramirez, Jose A. (Pronoun: he/him)
Wed, June 3	10	4:57: PM	Venkatesha Murthy, Veekshith (Pref: Veekshith, Pronoun: he, him)
Wed, June 3	10	4:59: PM	Handwerger, Adam L.
Wed, June 3	10	5:01: PM	Akula, Ranjith (Pronoun: he/him)
Wed, June 3	10	5:03: PM	Mickelson, Christian I. (Pref: Christian, Pronoun: He/Him)
Wed, June 3	10	5:05: PM	Shamszadeh, Navid (Pref: Navid, Pronoun: he/him)
Wed, June 3	10	5:07: PM	De Los Santos Razo, Jorge J. (Pronoun: He/Hims)